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Cognitive load and the effectiveness of distraction for acute pain in children

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Abstract

Background: Distraction tasks that place continuous, high demand on executive resources have been shown to reduce pain intensity and pain unpleasantness ratings in some healthy adult samples. We examined the effects of a high-demand 'working memory' 1-back task compared to a low-demand 'motor control' task on pain intensity and unpleasantness ratings in healthy children. Additionally, dispositional mindfulness was examined to explore the mechanisms of distraction on the affective processing of pain.

Methods: Fifty-seven children (9–13 years old) experienced three randomly presented heat levels (not painful, slightly painful, moderately painful) during two distraction conditions involving different levels of cognitive load (a high load 'working memory' task and a low load 'motor' control task) in counter-balanced order. Children completed measures of dispositional mindfulness, and attentional control and emotional control.

Results: As predicted, children's pain intensity and pain unpleasantness ratings were lower in the high load condition compared to the low load condition. These differences were amplified in the moderately painful heat trials. In contrast with predictions, dispositional mindfulness did not significantly predict the effectiveness of distraction. Dispositional mindfulness was significantly related to measures of children's attentional and emotional control abilities; however, an exploratory serial mediation model did not produce significant indirect or overall effects to suggest a strong influence of mindfulness on the effectiveness of distraction.

Conclusions: Results demonstrate that distraction that places higher demand on executive resources is more effective for acute pain management for children. Further research is needed to explore cognitive and affective moderators of the effectiveness of distraction for children.

Significance: This study is one of the first to demonstrate that working-memory engagement can attenuate pain intensity and pain unpleasantness in children aged 9–13. The findings suggest that distraction tasks used in clinical settings for moderately painful medical procedures may benefit more children if they are adequately demanding of cognitive resources.

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1 | BACKGROUND

Distraction is a widely studied intervention that has significant evidence to support its effectiveness in paediatric and adult acute pain management (Buhle & Wager, 2010; Dahlquist, Pendley, et al., 2002 Dahlquist, Weiss, et al., 2002; Legrain et al., 2013). Distraction is thought to function by limiting the finite attentional resources available for pain stimuli to capture (McCaul & Malott, 1984). Therefore, distraction tasks that require effortful use of attentional resources (e.g., working memory) use attentional resources that would otherwise be allocated to processing pain information, resulting in less pain perception.

In adults, working memory engagement has been shown in some samples to modulate cortical processing of painful stimuli such that nociceptive stimuli are not able to capture attention during task execution (Legrain et al., 2013), and reduce pain intensity (Buhle & Wager, 2010), albeit not in all studies (Seminowicz & Davis, 2007; Van Ryckeghem et al., 2018). One study to date has demonstrated the effectiveness of working memory engagement with children (Dahlquist, et al., 2019) using cold pressor pain tolerance as an outcome. The present study extends the study of working memory engagement for pain attenuation in children by using a different pain stimulus (heat) and by examining pain intensity and unpleasantness ratings.

Acute pain research often evaluates the effects of intervention on pain intensity ratings, which may neglect the nuances of affective aspects of pain. Pain intensity and pain unpleasantness are thought to reflect different processes in the pain experience (i.e., sensory vs. affective) (Miron et al., 1989), and activation of different brain regions involved in pain processing (Bentley et al., 2004; Kenntner-Mabiala et al., 2008; Rainville et al., 1997; Villemure & Bushnell, 2009). However, the relative effects of distraction on sensory and affective aspects of pain processing in children are not well-understood.

Finally, it is important to identify moderators of the effectiveness of distraction, as distraction is not equally effective for all individuals (Verhoeven et al., 2012). Specific executive functions that are considered critical for the effective utilization of distraction include the ability to: (a) sustain attention to the distraction task by easily shifting attention from painful stimuli; and (b) regulate emotional responses to pain stimuli (Moore et al., 2012). The current study explores these executive skills as potential moderators of distraction effectiveness.

Dispositional mindfulness – i.e., the *overall tendency* to non-judgmentally attend to present experiences and sensations – has been associated with better emotion regulation (Brown et al., 2007) and less negative emotion associated with pain (Lyvers et al., 2014; Teper & Inzlicht, 2013) in adults. Individuals who are skilled in attentional control, and early and implicit emotion regulation, may be better able attend to a distractor, in turn increasing the effectiveness of the distractor and reducing pain perception.

We hypothesized that all children would benefit from a 'high-load' working memory distraction intervention, reporting lower pain intensity and unpleasantness relative to a 'low-load' control condition. We also explored whether children higher in dispositional mindfulness and children with better attentional and emotional control executive skills would benefit the most from distraction.

2 | METHODS

2.1 | Participants

Those recruited for participation were 67 healthy children between the ages of 9 and 13 years from the University of Maryland Baltimore County Summer Day Camp and from the surrounding community. Parents who saw a flyer in the community contacted the lab to schedule an experimental session if their child was eligible. Parents and children attending the day camp were approached by study recruiters, provided study materials, consent forms, and questionnaires. Upon return of consent forms and questionnaires, children attending camp were scheduled to participate during or after camp. To be eligible to participate, children could not have an existing health condition (e.g., circulation problems, chronic pain, hearing or vision problems) and both parent and child had to be able to read or understand English. No children were excluded from participation due to not meeting these criteria.

Ten children were excluded from analyses for: computer malfunctions (2), lack of comprehension of experimental tasks (2), inconsistent baseline pain ratings (2) and missing data (4). The final sample included 57 children (*M* age =10.68 years, *SD* =0.91, 51% male). According to parent report, the sample was 63% White, 16% Black, 14% Biracial, 2% Asian or Pacific Islander, and 5% race not reported. The majority (82%) of boys were classified as pre-pubertal, whereas the majority (52%) of girls were classified as early. No participants were eliminated from analyses based on precocious puberty. See Table 1 for demographics and correlations with study variables.

2.2 | Measures

2.2.1 | Demographics and pubertal screening

Parents completed a demographic questionnaire that included information regarding their child's age, gender, and race. Parents also completed the parent version of the Pubertal Development Scale (Menseh et al., 2013), which classifies pubertal development based on parent report of external

TABLE 1 Correlation Matrix of Demographic Variables, Pain Rating Change Scores and Serial Mediation Variables

Variable	*N*	M	*SD*	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1. Child age	57	10.68	0.91	1.0															
2. Gender				0.22	1.0														
3. Pubertal status[a]				0.44**	0.61	1.0													
4. Race				−0.19	0.13	0.13	1.0												
Pain intensity change																			
5. Not painful	56	0.01	1.22	−0.34	−10	0.01	0.01	1.0											
6. Slightly painful	57	−0.37	1.30	0.14	−0.16	−0.08	−0.04	0.50**	1.0										
7. Moderately painful	57	−0.74	1.34	−0.11	−0.05	0.02	−0.08	0.60**	0.54**	1.0									
Pain unpleasantness change																			
8 Not painful	56	0.03	0.91	−0.17	0.02	0.04	−0.08	0.71**	0.53**	0.44**	1.0								
9. Slightly painful	57	−0.09	1.37	−0.05	−0.17	−0.17	0.03	0.48**	0.80**	0.31**	0.62**	1.0							
10. Moderately painful	57	−0.46	1.30	−0.17	−0.01	0.01	0.01	0.49**	0.42**	0.69**	0.55**	0.54**	1.0						
11. Mindfulness[b]	57	25.68	6.14	−0.18	0.12	−0.06	−0.09	−0.07	−0.12	0.01	−0.07	−0.17	0.11	1.0					
12. Attentional control[c]	57	25.05	4.67	−0.17	−0.03	0.03	−0.08	−0.12	−0.08	0.18	−0.13	−0.21	0.13	0.67**	1.0				
13. Emotional control[d]	57	10.98	3.16	0.05	0.01	0.21	0.17	0.06	−0.05	0.03	0.13	0.11	0.04	−0.36**	−0.21	1.0			
Task performance																			
14. Not painful	56	−0.07	1.47	0.16	0.35**	0.26	−0.09	−0.22	−0.39**	−0.15	−0.23	−0.42	−0.19	0.19	0.25	−0.06	1.0		
15. Slightly painful	57	−0.06	1.46	0.15	0.45**	0.26	−0.04	−0.13	−0.41**	−0.20	−0.17	−0.41	−0.19	0.15	0.15	−0.09	0.85**	1.0	
16. Moderately painful	57	−0.02	1.44	0.09	0.47**	0.38**	−0.04	−0.14	−0.36**	−0.18	−0.22	−0.40	−0.23	0.23	−0.27	−0.14	0.85**	0.85**	1.0

[a]3 participants declined to report pubertal status

[b]The Child and Adolescent Mindfulness Questionnaire (CAMM)

[c]The Attentional Control Scale for Children (ACS-C) Focus subscale

[d]The Behavior Rating Inventory of Executive Function- Second Edition (BRIEF-II) Emotional Control subscale

**Indicates statistical significance at $p < .01$.

puberty indicators. Although the research on puberty and pain remains inconclusive, some acute and chronic pain research has found that early puberty affects pain, particularly for girls (Janssens et al., 2011; Kløven et al., 2017). No children met the exclusionary criterion of precocious pubertal development.

2.2.2 | Attentional control scale for children (ACS-C)

The ACS-C was used to measure the construct of attentional control, specifically to capture children's self-reported ability to focus and shift their attention in every day, naturalistic settings (e.g., school). The ACS-C is a child adaptation of the adult version of the 20-item Attentional Control Scale (ACS) (Muris et al., 2004). Designed for children ages 8–13, it is composed of 20 items measuring two types of attentional control: Focus (e.g., "When concentrating, I do not notice what happens around me") and Shift (e.g., "I can easily write or read while I am talking on the phone"). Items are scored on a 4-point Likert scale from 0 = *never*, 1 = *sometimes*, 2 = *often*, and 3 = *always*. Nine items were reverse scored. After reverse scoring, higher scores indicate higher levels of attentional control. The developers of the ACS-C report adequate internal consistency (αs range from 0.72 to 0.75) as well as positive correlations with perceived control ($r = 0.22$) and negative correlations with trait anxiety ($r = -0.38$) and neuroticism ($r = -0.40$) as measured by the Screen for Child Anxiety (SCARED) suggesting adequate construct validity (Muris et al., 2004). In adults, ACS scores predict better performance on tasks involving goal-directed focus and inhibition of prepotent responses (Judah et al., 2014).

Adequate internal consistency for the 9-item Focus subscale scores was demonstrated in the present study, with Cronbach's α of 0.75. However, the Shift subscale revealed low internal consistency with the six items ($\alpha = 0.33$) that have been shown to load on the subscale in validation studies. The reliability of the scale was improved in this sample by removing item 14 ($\alpha = 0.46$), yet was insufficient for use in analyses. Therefore, only the Focus subscale was used in analyses.

2.2.3 | Behaviour rating inventory of executive function (BRIEF-2©)

The BRIEF-2 Parent Form (Gioia et al., 2015) was used to measure children's level of executive functioning. The BRIEF-2 is an 86-item measure that assesses eight clinical executive functioning subscales and has demonstrated adequate reliability and validity in a standardization sample of 1,400 children aged 5–18 years (Gioia et al., 2015).

For the purposes of this study the raw scores from the Emotional Control subscale were used, to estimate the level of the child's skills, rather than a comparison to clinical norms (i.e., standardized t-scores). The Emotional Control subscale consists of eight items that measure children's ability to modulate emotional responses (e.g., "small events trigger big reactions," and "mood is easily influenced by the situation"); lower scores indicate better emotional control. The developers of the BRIEF-2 reported good internal consistency (Cronbach's $\alpha = 0.89$) and adequate 3-week test–retest reliability ($r = 0.79$) for the Emotional Control subscale. Cronbach's α for the Emotional Control subscale in the present study was 0.87.

2.2.4 | Child and adolescent mindfulness measure (CAMM)

Children completed the 10-item CAMM (Greco et al., 2011) to assess present-moment awareness and non-judgmental, non-avoidant responses to thoughts and feelings. The CAMM was administered in interview format. Children were asked to rate how often each item is true for them using a 5-point Likert scale anchored 0 = *never true,* and 4 = *always true*. Possible scores range from 0 to 40, with higher scores indicating higher levels of dispositional mindfulness.

The CAMM has shown adequate internal consistency (between 0.70 and 0.80) across multiple samples and a one-dimensional factor structure (de Bruin et al., 2014; Kuby et al., 2015). CAMM scores have been found to negatively correlate with child-reported somatic complaints, internalizing symptoms, thought suppression, and psychological inflexibility (Greco et al., 2011), as well as self-reported negative affect and worry (Kuby et al., 2015), and to be positively related to healthy self-regulation (de Bruin et al., 2014). Adequate internal consistency for the CAMM was found in the present study, with Cronbach's α of 0.76.

2.2.5 | Pain ratings

Children's subjective ratings of pain were measured using an 11-point visual analogue scale (VAS) displayed horizontally on a computer screen. Participants were used the "1" and "2" keys on a numerical keypad to move a cursor left or right along the line to indicate their rating. For *pain intensity*, participants responded to the cue, "How painful?" by moving the cursor along a VAS anchored 0 "*no pain at all"* to 10 "*the worse pain I have ever felt*." Children's subjective ratings of *pain unpleasantness* were measured the same way, responding to the cue, "How unpleasant?" using a VAS anchored 0 "*not at all unpleasant"* to 10 "*the most unpleasant pain I have ever felt*." No numbers appeared between the descriptors on the scales. The starting point of the cursor was located at the far left of the screen, on the number 0, for all ratings.

2.3 | Equipment

2.3.1 | Pain stimulus

Heat was delivered via the FDA-approved Medoc Pathways system (MEDOC Ltd. Systems, n.d.) using a 16 × 16 mm Advanced Thermal Stimulator (ATS) contact thermode. For safety, a maximum temperature limit of 50°C was strictly enforced for trials involving 1 s exposure (i.e., threshold and tolerance trials) which is consistent with the majority of thermal pain research studies with adults and children (Buhle & Wager, 2010; Lu et al., 2007). A maximum temperature limit of 49°C was enforced for trials involving 6–11 s of exposure. A standard computer mouse with two buttons was used for participant responses to terminate the heat stimulus on the threshold and tolerance calibration trials.

2.3.2 | Computer

Experimental tasks were generated using E-prime 2.0 software (Schneider et al., 2002), a Dell Latitude E7240 laptop with an Intel® Core™ i7-5600 U 2.60GHz CPU, and a 92 cm LCD flat panel display. A standard numerical keypad with the "1" key labelled "Yes" and the "2" key labelled "No" was used for participant responses on the experimental tasks.

2.4 | Setting

Experimental procedures occurred in a soundproof room (approximately 3 m by 3 m) maintained between 23 and 24°C. Children sat in a comfortable armchair. One experimenter sat on the dominant-hand side of the child. The other experimenter stood directly behind the child to operate the MEDOC system and to adjust the thermode on the child's arm. A small desk was placed on the child's dominant side to allow for placement of the response mouse or keypad. All experimental stimuli were presented on the computer monitor approximately 60 cm in front of the child.

2.5 | Procedure

All procedures followed were in accordance with the ethical standards of the responsible committee on human experimentation (institutional and national) and with the Helsinki Declarations. All procedures were approved by the University of Maryland, Baltimore County Institutional Review Board. Undergraduate and graduate students with experience working with children served as experimenters.

2.5.1 | Parental consent, child assent and CAMM administration

Parents and children were informed that the heat delivered during the experiment might cause the child's skin to appear pink for a short period of time following participation, but that safety settings on the equipment would ensure that children could not obtain a burn from the thermode. Assent forms were read out loud, with the child following along with their own copy. Experimenters then answered questions and assured children that they could choose to not be in the study or to stop participating at any time, and obtained informed assent. The experimenter then read the CAMM aloud to children and recorded responses.

2.5.2 | Thermal stimulus calibration

To ensure that individualized suprathreshold temperatures that related to the child's personal pain threshold and tolerance were used in the subsequent experimental trials, participants completed a series of thermal stimulus calibration trials. The following values were calibrated: warmth threshold, pain tolerance and pain threshold. Pain tolerance was assessed prior to pain threshold to allow participants to experience the full range of painful heat stimuli, with the goal of reducing anticipatory anxiety that might influence the accuracy of pain threshold data.

The experimenter directed the child to remain seated in a chair for the duration of the experiment, with their arm on top of a pillow placed on the armrest or their lap with the palm facing upward. The experimenter placed the thermode on the volar surface of the child's non-dominant arm halfway between their wrist and their elbow. The thermode was comfortably secured on their arm using a Velcro strap. The experimenter positioned the MEDOC response mouse under the child's dominant hand. During all stimulus calibration trials the thermode temperature started at 32°C, increased at a rate of 1°C per second, up at a maximum of 50°C, and returned to baseline (32°C) at a rate of 8°C per second following a response (clicking the mouse). The temperature then remained at baseline for 8 s prior to the onset of the subsequent trial.

2.5.3 | Warmth threshold

A set of three stimulus calibration trials were conducted, during which children were instructed to click any button on the mouse when they first noticed that the thermode changed from 'room temperature' to 'warm.' The average temperature of the second and third trials was calculated to determine the child's *warmth threshold*. Pearson correlations revealed a strong association between the second and third trials ($r = 0.64$).

2.5.4 | Pain tolerance

During the second set of three stimulus calibration trials, children were instructed to click any button of the mouse when they first noticed that the thermode became 'too uncomfortable' or 'too painful' to continue. The average temperature of the second and third trials was calculated to determine the child's *pain tolerance*. The second and third trial pain tolerance scores were highly correlated ($r = 0.87$).

2.5.5 | Pain threshold

During the third and final set of three stimulus calibration trials, children were instructed to click any button on the mouse when they 'first noticed' the thermode change from 'warm' to 'just painful.' The average of the last two trials served as the child's *pain threshold*. Pearson correlations revealed a strong association between the second and third trials ($r = 0.89$).

2.5.6 | Reliability series

Following the stimulus calibration trials, the experimenter tested the reliability of the child's subjective pain ratings to longer duration heat stimuli. Each stimulus lasted 13 s, with a temperature ramp rate between 1°C per second and 3.2°C per second, depending on the target peak temperature. Two trials at each of the following three heat levels were presented in a fixed order: (a) 37°C ('not painful'), which is a temperature that does not stimulate nociceptors (Dubin and Patapoutian, 2010); (b) temperature equal to the child's average pain threshold ('slightly painful'); and (c) temperature 1°C higher than pain threshold ('moderately painful'). Heat stimuli were presented in the following order for all participants to reduce expectancy effects: 'slightly painful', 'moderately painful', 'not painful', 'moderately painful', 'not painful,' 'slightly painful.'

Following each reliability trial, the child rated pain intensity and unpleasantness. If needed, the experimenter adjusted the heat levels and repeated reliability trials to obtain pain intensity ratings between 5 and 7 for the 'moderately painful' heat level using a standardized protocol. Heat levels were adjusted for 39% of children to reach desired pain intensity ratings. Final program temperatures ranged from 40°C to 48°C for 'moderately painful' stimuli (*M* temp =43.50, *SD* =2.67) and from 39°C to 47°C for 'slightly painful' stimuli (*M* temp =42.50, *SD* =2.67). During reliability trials, participant's variability within their own two ratings for 'moderately painful' heat levels was significantly related to their variability within the control condition for 'moderately painful' heat stimuli ($r = 0.30$), suggesting that children tended to be consistent within themselves within the 'moderately painful' heat trials. However, the correlations between reliability and control trials were lower for the 'not painful' and 'slightly painful' trials $r = 0.15$, and $r = 0.17$, respectively.) *Experimental design*. A modification of the experimental procedures used by Buhle & Wager (Buhle & Wager, 2010) was employed. Children completed two blocks of nine trials, which lasted 50 s per trial. In each block they experienced the three individually determined heat stimuli (not painful, slightly painful, moderately painful) in the same fixed order. During one block they performed the 1-back task; during the other they performed the control task. Order of experimental and control blocks was counterbalanced by child age and sex. To control for potential attrition effects on order counterbalancing, the experimental order that was assigned to the participant who dropped out was replaced in the pool of randomization options prior to randomization of the subsequent participant; thus, the number of participants in each order was equal, (n = 28). Pain intensity and unpleasantness ratings were obtained after each trial. Average pain ratings were calculated for each heat level within each block.

2.5.7 | 1-*back (high load 'working memory')* condition

During each 1-back trial, participants saw a series of 22 numerals, each presented for 840 ms. A randomly-generated 1,000 millisecond serial letter mask appeared between each numeral presentation. Experimental stimuli consisted of 120-pt Times New Roman font white upper case letters and numerals presented on a black background. Participants were asked to report if the number presented was the same as the number immediately preceding it by pressing the '*yes*' or '*no*' button on the keypad. Of the 22 stimuli in each trial, 30% were the same as the preceding stimulus.

As an estimate of task performance that adjusts for response bias and careless/random responding, d-prime (*d'*) scores were calculated for each trial by subtracting the standardized false alarm (z) score from the standardized hits (z) score. Possible *d'* scores range from −4.65 to 4.65.

2.5.8 | Control condition

To control for the motor movement involved in indicating a response and the act of attending to visual stimuli, children viewed a series of 22 random numerals (integers 1 through 9), each of which was presented for 840 millisecond and followed by a 1,000 millisecond serial letter mask in the same manner as in 1-back condition. Children were instructed to press the "*yes*" key on the keypad to indicate ('*yes, I see a number*') every time a number appeared.

2.5.9 | Task training

Children were trained on the control and 1-back tasks to ensure that they understood the task demands prior to the start of the experimental trials. Following demonstration of their comprehension of the basic rules of the task, children completed a mock trial of the same duration as the actual experimental trial, but without heat stimuli, with the goal of reaching 80% accuracy. Children completed as many trials as it took to achieve 80% accuracy up to a maximum of three mock trials.

To increase motivation and task persistence, children were told that they could earn up to five "coins" (i.e., pretend money) that they could trade in for prizes at the end of the study depending on their performance on the task. For both tasks, children earned the five coins at the end of the mock trial in which they reached 80% accuracy, or after the third mock trial, regardless of accuracy.

Three children required a second trial of the control task to achieve 80% accuracy. On the 1-back, 80% accuracy was demonstrated by 12 children on the first trial, 22 children on the second trial and four children on the third trial. Accuracy scores of the remaining 19 children ranged from 50% to 77.27%, with a mean 65.79%. These participants all showed comprehension of the 1-back instructions and persistent effort on the task, and therefore their 1-back data were included in analyses.

2.5.10 | Experimental trials

The children were notified at the start of each block whether the subsequent trials required performance of the 1-back task or the control task. Each trial began with a 4-s fixation cross to orient the child to the location on the screen where the experimental stimuli would appear. At 26 s a warning tone sounded and one of the three heat levels was delivered. At 39 s, the temperature returned to baseline and the experimental task ended. Children immediately rated pain intensity and pain unpleasantness on the VAS scales presented on the screen. Children were told that they could earn up to five "coins" (i.e., pretend money) if they "did well" on the task to provide a standardized motivator designed to encourage attendance to both tasks. All children earned five coins at the end of each experimental block. Children completed the ACS-C between the two experimental blocks. Throughout experimental blocks, research assistants were trained to monitor participant's response patterns, ensure that their eye gaze was fixed on the screen for each trial, and monitor that responding was concurrent with visual presentations. The percentage of correct responses for each trial was recorded using E-prime software. Across trials in the control condition, the mean percent correct responses ranged from 96.25% to 97.53%. Across trials in the 1-back condition, the mean percent correct responses ranged from 79.74% to 83.65%. Descriptive data for each percentage of correct responses by trial are presented in the Table S1.

The entire experiment lasted approximately 1 hr. Children received $10.00 and a small toy (exchanged for "coins") for their participation. Parents whose children did not attend summer camp received $10.00 to cover transportation expenses.

2.6 | Data analytic approach

Pain intensity and pain unpleasantness change scores were calculated by subtracting average pain ratings in the control condition from average pain ratings in the 1-back condition separately for each of the three heat levels. Negative change scores indicated lower pain ratings during 1-back distraction (compared to control); positive change scores indicated that pain intensity increased during the 1-back condition relative to the control condition (i.e., the 1-back distraction was not effective).

Descriptive analyses were conducted to examine the normalcy of distributions to ensure that all variables met assumptions for analysis. Correlational analyses were conducted to examine the relations between child age, gender, and pubertal status and the pain outcome variables to identify potential covariates. A repeated measures ANOVA was conducted to examine the mean differences in pain ratings for each of the three heat levels in the control condition to confirm that pain ratings increased from not painful stimuli to slightly painful stimuli to moderately painful stimuli. Separate repeated measures $2 \times 3 \times 2$ (experimental condition by heat by order) analyses of variance ANOVAs were conducted to examine order effects.

Separate 2×3 (experimental condition by heat) analyses of variance (ANOVAs) were conducted to examine the effects of high load distraction and heat level on pain intensity and pain unpleasantness ratings. Post hoc tests were conducted to compare mean differences between experimental conditions at each of the heat levels. For all post-hoc analyses, Cohen's d_{rm} effect sizes are reported to statistically control for correlations between the repeated measurements (Lakens, 2013).

To examine dispositional mindfulness as a moderator of response to distraction, separate 2×3 (experimental condition by heat) ANCOVAs with dispositional mindfulness entered as a continuous covariate were conducted for pain intensity and pain unpleasantness ratings.

The PROCESS macro for SPSS (Hayes, 2013) was used to test an exploratory analysis to evaluate the expected direct and indirect effects of emotional control, attentional control and 1-back task performance on the relation between dispositional mindfulness and distraction effectiveness (i.e., pain unpleasantness change scores).. The serial mediation model was evaluated by examining the bootstrapped 95% confidence intervals for direct and indirect effects of (a) emotional control; (b) attentional control and (c) 1-back task performance on the relation between dispositional mindfulness and distraction effectiveness.

2.7 | Power analyses

To determine the number of participants required to detect each expected effect a priori power analyses were conducted using *G*Power 3.1* (Faul, Erdfelder, Buchner, & Lang, 2009) and MedPower for the indirect and direct effects within the full mediation analysis (Kenny, 2017) (Kenny, 2017). To provide the minimal detectable effect sizes for each analysis given the final sample size of $n = 57$, sensitivity power analyses are also provided using *G*Power 3.1* (Lakens, 2021). All power analyses were conducted using the assumptions of a one-tailed test with an alpha of 0.05. For the main effect of cognitive load on distraction effectiveness, past research has shown effects ranging from no effect (Seminowicz & Davis, 2007) to large effects (Bantick et al., 2002; Buhle & Wager, 2010; Wiech et al., 2005). For the current study, the main effects of cognitive load on pain ratings were powered at 0.80 to detect a medium effect (Cohen's $f = 0.25$) with a sample of 27 children. The minimal detectable effect size with a sample size of 57 children is Cohen's $f = 0.13$. With regard to the hypothesized effects of dispositional mindfulness, previous research has shown no effect to medium effects between dispositional mindfulness and pain ratings (Perlman et al., 2010; Prins et al., 2014; Zeidan et al., 2010). The relation between dispositional mindfulness and pain ratings were powered at 0.80 to detect a medium effect ($r = 0.30$) with a sample of 66 children. The moderation effects of dispositional mindfulness between the effects of cognitive load on pain ratings were powered at 0.80 to detect a medium effect with 157 children. The minimal detectable effect size with a sample size of 57 children is Cohen's $f = 0.45$. Past research shows medium to large effects between dispositional mindfulness and emotional control (Brown et al., 2013; Teper & Inzlicht, 2013), and dispositional mindfulness and attentional control (Brown et al., 2013; Zylowska et al., 2008). The serial mediation was powered at 0.80 to detect a small to medium indirect effect of mindfulness on pain unpleasantness change with 168 children, with smaller sample sizes needed to detect anticipated indirect mediation effects within the larger model for emotional control, attentional control and task performance, respectively ($n = 45$, $n = 84$ and, $n = 212$), as well as direct effects.[1]

3 | RESULTS

3.1 | Preliminary analyses

Pearson correlations revealed no significant relations between child age, gender, pubertal status and pain change scores, $p > 0.05$. Therefore, these variables were not included as covariates in subsequent analyses. Tests for skewness and kurtosis suggested that no transformations were necessary.

3.1.1 | Heat stimulus manipulation check

A repeated measures ANOVA, with heat stimulus (not painful, slightly painful, moderately painful) as the within-subjects factor and pain intensity ratings in the control condition as the dependent variable revealed a significant main effect of heat, $F(2,112) = 81.42, p < 0.001$. Post-hoc t tests indicated that participants rated pain intensity of the three heat stimuli in the expected manner. Specifically, pain intensity ratings were significantly lower in the not painful heat trials ($M = 2.61$, $SD = 1.55$) compared to the slightly painful ($M = 3.79$, $SD = 1.75$) and moderately painful heat trials ($M = 4.92$, $SD = 2.00$), ($t[56] = -6.63$, $p < 0.001$, Cohen's $d_{rm} = 1.07$, and $t(56) = -11.10$, $p < 0.001$, Cohen's $d_{rm} = 1.73$, respectively). Additionally, pain intensity ratings were significantly lower in the slightly painful heat trials compared to the moderately painful heat trials, $t(56) = -7.69$, $p < 0.001$, Cohen's $d_{rm} = 0.93$.

A repeated measures ANOVA, with heat stimulus (not painful, slightly painful, moderately painful) as the within-subjects factor and pain unpleasantness ratings as the dependent variable also revealed a significant main effect of heat, $F(2,112) = 71.33$, $p < 0.001$. Post-hoc t tests indicated that participants also rated pain unpleasantness of the three heat stimuli in the expected manner. Pain unpleasantness ratings were significantly lower in the not painful heat trials ($M = 2.25$, $SD = 1.41$) compared to the slightly painful ($M = 3.33$, $SD = 1.87$), and moderately painful heat trials ($M = 4.43$, $SD = 2.03$), ($t(56) = -6.56$, $p < 0.001$, Cohen's $d_{rm} = 1.08$, and $t(56) = -10.29$, $p < 0.001$, Cohen's $d_{rm} = 1.77$, respectively), and significantly lower in slightly painful heat trials compared to moderately painful heat trials, $t(56) = -6.59$, $p < 0.001$, Cohen's $d_{rm} = 0.91$.

3.1.2 | Order effects

Separate repeated measures $2 \times 3 \times 2$ (experimental condition by heat by order) ANOVAs did not reveal a significant interaction of order, for pain intensity as the dependent variable, $F(2,108) = 0.43$, $p = 0.654$, Cohen's $f = 0.09$ or for pain unpleasantness ratings as the dependent variable, $F(2,108) = 1.03$, $p = 0.360$, Cohen's $f = 0.14$.

3.2 | Cognitive load and distraction effectiveness

A repeated measures 2 × 3 (experimental condition by heat) ANOVA, with experimental condition (control vs. 1-back) as one within-subjects factor, heat stimulus (not painful, slightly painful, moderately painful) as the second within-subjects factor and pain intensity as the dependent variable revealed a significant condition by heat interaction, $F(2,55) = 10.12$, $p < 0.001$ (see Table 2). Post-hoc t tests showed that pain intensity ratings were significantly lower in the 1-back condition compared to the control condition in both slightly painful ($t[56] = 2.14$, $p = 0.036$, Cohen's $d_{rm} = 0.28$) and moderately painful $t(56) = 4.16$, $p < 0.001$, Cohen's $d_{rm} = 0.51$) trials. As would be expected, experimental condition had no effect on pain intensity ratings in the not painful heat trials, $t(56) = -0.07$, $p = 0.942$, Cohen's $d_{rm} = 0.01$). Figure 1 depicts mean pain intensity ratings for each heat stimulus in the control and 1-back conditions.

A repeated measures 2 × 3 (experimental condition by heat) ANOVA with experimental condition (control vs. 1-back) as one within-subjects factor heat stimulus (not painful, slightly painful, moderately painful) as the second within-subjects factor, and pain unpleasantness as the dependent variable, also revealed a significant condition by heat interaction, $F(2,55) = 4.71$, $p = 0.011$. Post-hoc t tests showed that 1-back distraction significantly reduced pain unpleasantness ratings compared to the control condition in moderately painful trials $t(56) = 2.64$, $p = 0.011$, $d_{rm} = 0.33$, but not in slightly painful trials $t(56) = 0.52$, $p = 0.608$, $d_{rm} = 0.07$ or in the not painful trials $t(56) = -25$, $p = 0.807$, $d_{rm} = 0.02$. Figure 2 depicts mean pain unpleasantness ratings for each heat stimulus in the control and 1-back conditions.

3.3 | Dispositional mindfulness and pain ratings

During the control condition, dispositional mindfulness was not significantly correlated with pain unpleasantness ratings for either the slightly painful or moderately painful trials

TABLE 2 Summary of 2 x 3 (Experimental Condition x Heat) and 2 x 3 x 2 (Experimental Condition x Heat x Dispositional Mindfulness) ANOVAs on Pain Ratings

Dependent Variable	Effect	F	p	η_p^2
Pain Intensity	Condition	6.30	0.015	0.10
	Heat	81.41	<0.001	0.60
	Condition x Heat	10.12	<0.001	0.16
	Condition	3.27	0.080	0.09
	Heat	58.25	<0.001	0.64
	Mindfulness	1.08	0.408	0.41
	Condition x Heat	6.67	0.003	0.17
	Condition x Mindfulness	0.53	0.940	0.26
	Heat x Mindfulness	0.88	0.672	0.37
	Condition x Heat x Mindfulness	0.90	0.649	0.37
Pain Unpleasantness	Condition	1.42	0.239	0.03
	Heat	69.07	<0.001	0.56
	Condition x Heat	4.71	0.011	0.08
	Condition	0.87	0.357	0.03
	Heat	60.68	<0.001	0.65
	Mindfulness	0.93	0.566	0.38
	Condition x Heat	3.26	0.045	0.09
	Condition x Mindfulness	0.64	0.864	0.30
	Heat x Mindfulness	1.18	0.270	0.44
	Condition x Heat x Mindfulness	0.99	0.507	0.40

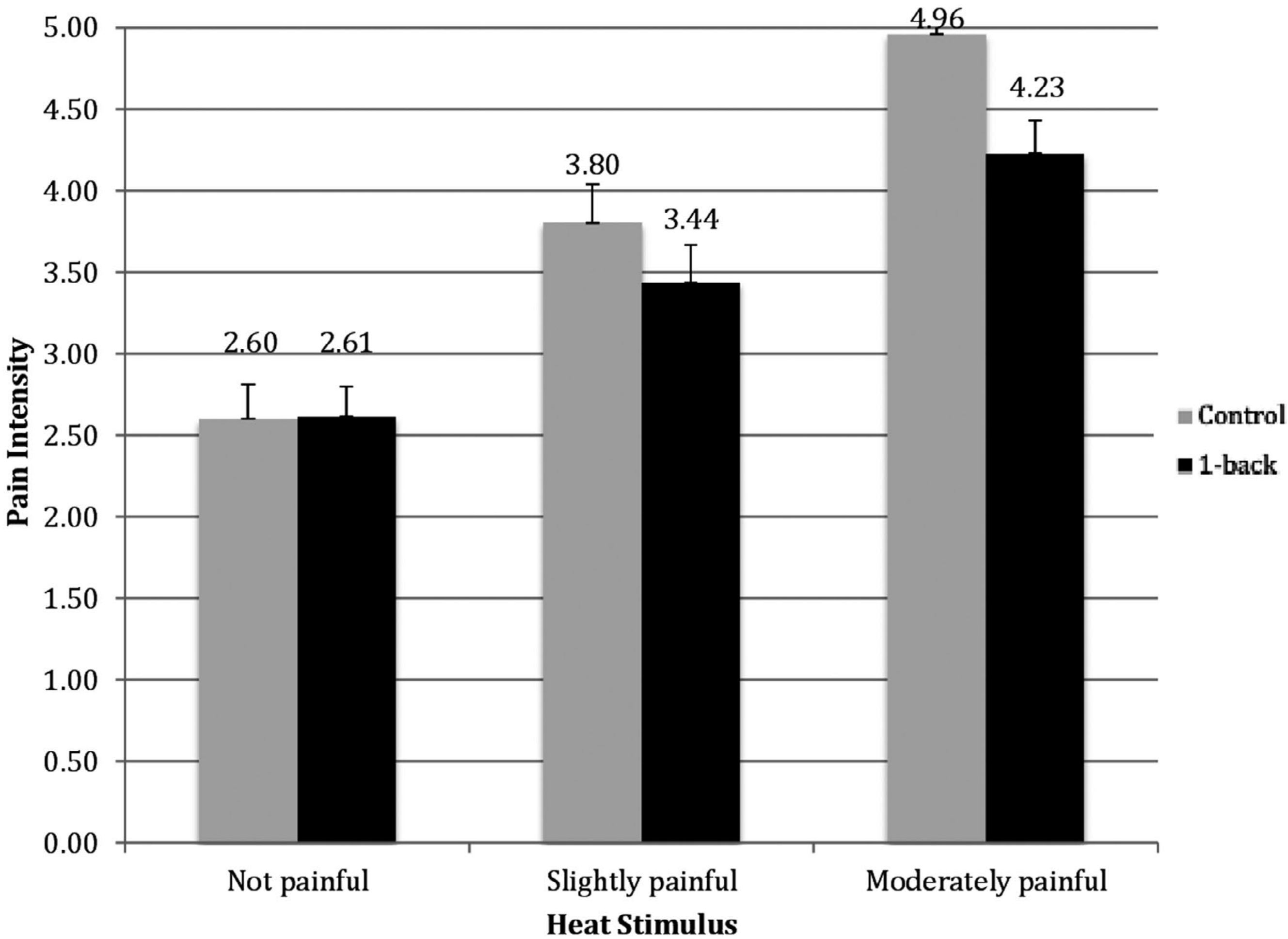

FIGURE 1 Mean pain intensity ratings during 1-back and control conditions. Error bars represent standard deviations

($r = 0.14$, $p = 0.296$, $r = 0.04$, $p = 0.766$, respectively). Similarly, dispositional mindfulness was not significantly correlated with pain intensity ratings for slightly painful or moderately painful trials ($r = 0.08$, $p = 0.576$, $r = 0$ 0.05, $p = 0.699$, respectively).

3.4 | Dispositional mindfulness as a moderator of distraction effectiveness

Separate 2 × 3(experimental condition by heat) repeated measures ANCOVAs, with mindfulness level entered as a continuous factor were conducted to examine the effect of dispositional mindfulness on the effects of high load distraction and heat level on pain intensity and pain unpleasantness ratings. However, the interaction between mindfulness and experimental condition was not significant at any of the heat levels for pain intensity or pain unpleasantness (see Table 2). Thus, mindfulness did not appear to moderate the effects of experimental condition on pain intensity or pain unpleasantness ratings.

3.5 | Serial mediation of the effect of dispositional mindfulness on change in pain unpleasantness

An exploratory serial mediation model tested the effect of dispositional mindfulness on distraction effectiveness (pain unpleasantness change from control to 1-back condition) in the moderately painful heat trials, through three predictors: emotional control, attentional control and 1-back task performance (*d'*). All indirect effects were tested using bootstrap confidence intervals with 5,000 samples. See Figure 3 for a visual depiction of the serial mediation model and pathways.

Dispositional mindfulness did not significantly predict distraction effectiveness after controlling for the other variables in the model, as evidenced by a non-significant total effect c = 0.02, $p = 0.401$. Within the model, dispositional mindfulness directly predicted emotional control, $a_1 = -0.17$, $p = 0.012$, as well as attentional control, $a_2 = 0.45$, $p < 0.001$. Mindfulness did not significantly predict 1-back task performance, $a_3 = 0.03$, $p = 0.520$.

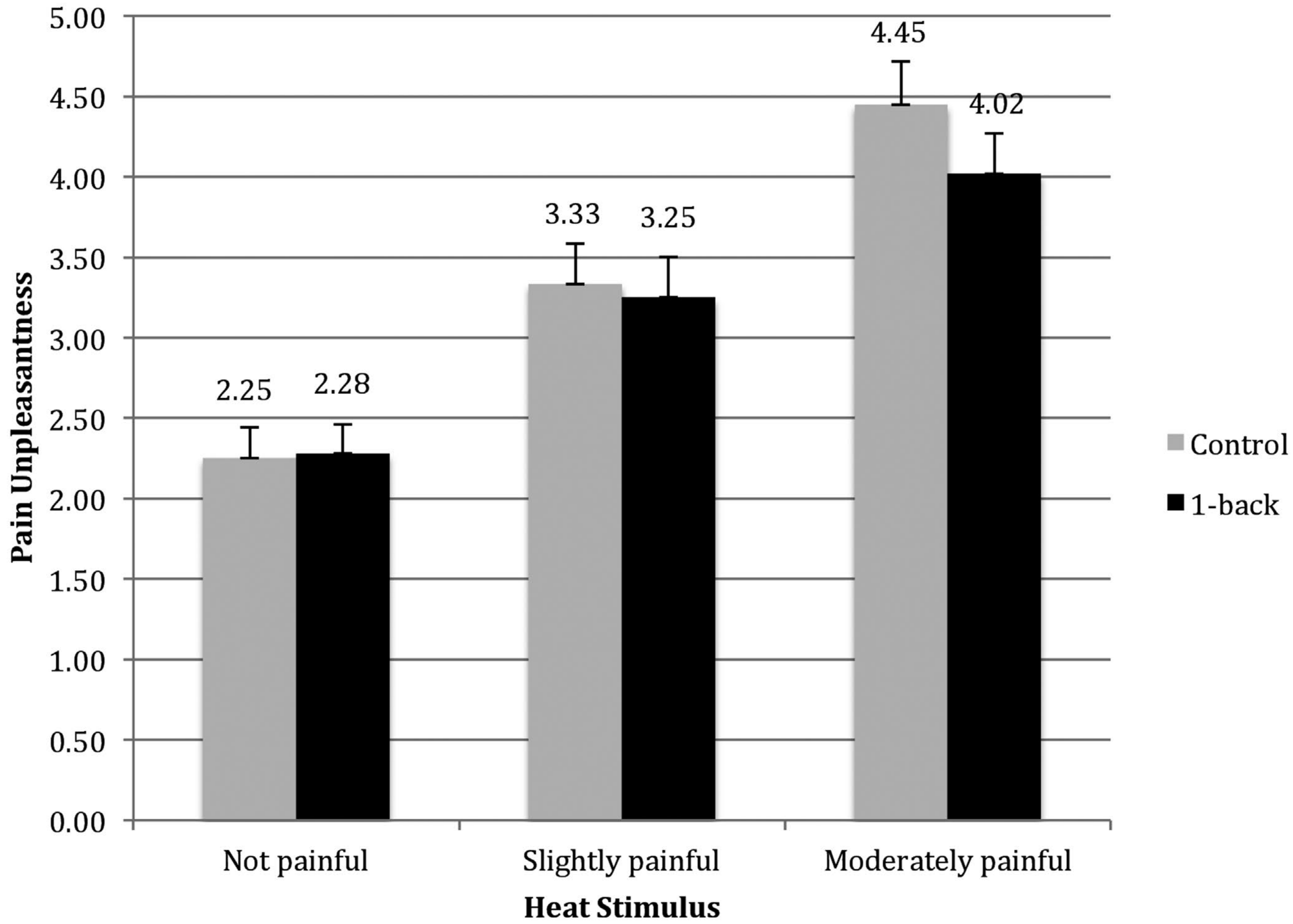

FIGURE 2 Mean pain unpleasantness ratings during 1-back and control conditions. Error bars represent standard deviations

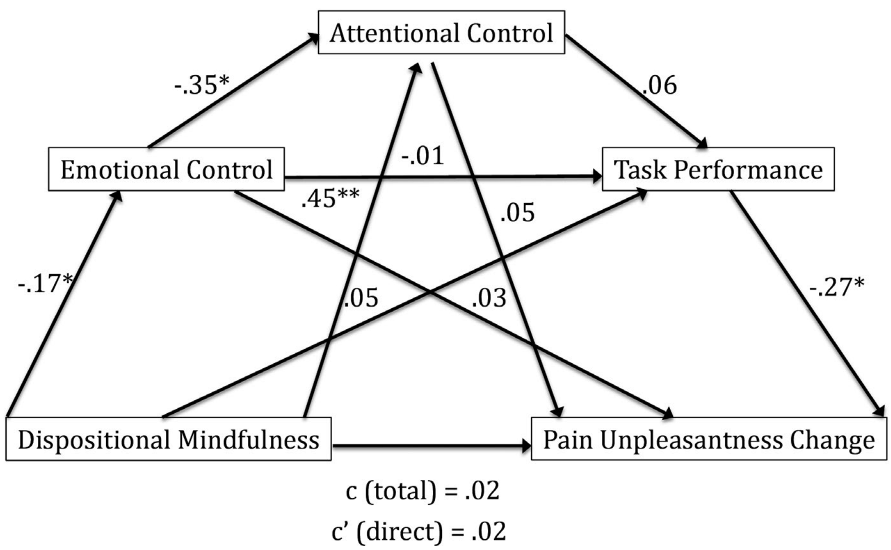

FIGURE 3 Serial mediation model of the effects of dispositional mindfulness on pain unpleasantness change. Direct and indirect effects

The indirect effects within the model revealed no significant mediating effects, indicating that emotional control, a_1 b_1 = −0.01, CI [−0.04, 0.01], attentional control, a_2 b_2 = 0.00, CI [−0.01, 0.01] and 1-back task performance a_3 b_3 = 0.00 CI [−0.04, 0.01] did not significantly mediate the relation between dispositional mindfulness and distraction effectiveness. Additionally, 1-back task performance directly predicted distraction effectiveness, after controlling for all other predictors, $b_3 = -0.27$, $p = 0.038$ and emotional control directly predicted attentional control, after controlling for dispositional mindfulness, $d_{21} = -0.35$, $p = 0.027$.

4 | DISCUSSION

This is the first study to show an effect of cognitive load on distraction effectiveness in children. The methodology closely replicated an experimental paradigm developed to isolate cognitive load as the only manipulated factor between experimental conditions (Buhle & Wager, 2010). As expected, the results were consistent with their findings with adults that the high load 'working memory' (1-back) distraction intervention was more effective for reducing pain intensity than the low load 'motor' control condition at both levels of painful heat (slightly painful and moderately painful).

A similar pattern was detected with regard to pain unpleasantness ratings. Specifically, high load distraction was more effective for reducing pain unpleasantness ratings at moderately painful heat levels than the low load control condition. The effects were smaller than the effects for pain intensity, suggesting that distraction may reduce the sensory aspects of pain processing to a greater degree than the more affective components of pain that are captured by pain unpleasantness ratings. It is possible that a task that manipulates affective states to a greater degree than a working memory task will have a greater impact on pain unpleasantness. However, further research is needed to determine whether a task that manipulates affect (e.g., a more enjoyable high-load task) might have a greater impact on pain unpleasantness, which would have implications for clinical use for acute pain management.

The current findings are consistent with the argument that pain and distraction compete for limited cognitive resources (Frankenstein et al., 2001; Veldhuijzen et al., 2006). The low load 'motor control' condition involved sustained visual attention, repetitive motor movements and responding, whereas the high load 1-back condition required additional working memory processing, making it more effective at competing for attentional resources (Eccleston, 1995). This is consistent with previous findings that some prefrontal cortical areas (i.e., dorsolateral prefrontal cortex) are only activated during high load tasks. However, there is mixed evidence in the literature regarding the effects of cognitive load on distraction effectiveness in adults (Seminowicz & Davis, 2007; D. M. V. Van Ryckeghem et al., 2018); additional research is needed to further explore potential moderators in children, including contextual and individual factors (e.g., motivation, affect, task qualities) (Birnie et al., 2017).

The benefit of cognitive load on perceived pain intensity increased as heat intensity increased, as evidenced by a larger effect size for moderately painful heat stimuli (0.41) than it for slightly painful heat stimuli (0.21). This finding is consistent with some previous adult research (Wiech et al., 2005) that found a large effect of cognitive load at higher levels of pain stimulation, and no differences between high load and low load distraction at low levels of pain stimulation. However, this finding is limited to pain stimuli that are below pain tolerance, "moderately painful" stimuli (i.e., between a 5 and 7 on the rating scale) as there is evidence to suggest that distraction may be less effective at higher pain intensity levels (e.g., pain tolerance) (Van Ryckeghem et al., 2012). In the current study, we selected pain stimuli in this range as it 1) reduces the risk of ceiling and floor effects, and 2) is tolerable to participants.

However, the amount of cognitive load that is necessary to interfere with pain processing in children remains unknown. In our own recently published study (Dahlquist, et al., 2019), we found no difference in effects on cold pressor pain tolerance between a 1-back task and a less effortful simple visual discrimination task (indicating when a numeral 5 appeared with "yes" and a numeral between 1–9 other than 5 with "no"). Both tasks improved pain tolerance relative to no distraction, suggesting that both tasks were effortful enough for children to interfere with pain processing. In our previous study, and the current study, we recorded participant's performance in all trials for tasks, which may shed some light on actual task engagement and relative task difficulty level. In the present study, mean correct responses indicate that the control condition was less difficult than the 1-back task. Comparative data of the two studies will be presented in a separate manuscript.

Although dispositional mindfulness demonstrated expected positive relations with attentional control and emotional control, contrary with prediction, dispositional mindfulness did not moderate the effectiveness of distraction. It is possible that mindfulness abilities were not yet adequately developed in this sample of 9- to 13-year-olds to be able to be applied to this context. It is also possible that the mindfulness measure used in the present study may not have been sensitive to pain contexts. A dispositional mindfulness measure that includes pain-related non-judgmental attention, or pain-related catastrophizing may show a more robust prediction of pain outcomes. Alternatively, children's actual daily exposure to and practice of mindfulness (Lyvers et al., 2014; Perlman et al., 2010) may prove to be a better predictor of responses to acute pain.

Finally, it may be the case that children who are high in mindfulness do not need or do not benefit from distraction. For example, in adults, participation in three days of mindfulness training showed increases in dispositional mindfulness as well as reduced pain sensitivity but did not predict benefit from an experimental math distraction task for reducing pain intensity ratings (Zeidan et al., 2010). Thus, children who naturally attend to pain more non-judgmentally may be more likely to benefit from actively attending, non-judgmentally, to pain stimuli to reduce its unpleasantness rather than attempting to distract from the pain. Although past research attempting to match children's coping style to acute pain interventions has been inconsistent, mindfulness was not examined in these studies (Piira et al., 2006).

Although not shown to moderate distraction effectiveness, mindfulness did demonstrate important theoretically relevant relations with factors thought to be associated with pain. For example, dispositional mindfulness was significantly related to emotional control. Parents of children who were higher on dispositional mindfulness reported that their children had fewer emotional control problems and better attentional control abilities than parents of children who were lower on dispositional mindfulness. These findings are in line with previous research in adult and child samples (Brown et al., 2007; Zylowska et al., 2008). Research with adults has suggested that mindfulness likely improves emotion regulation abilities by way of reducing the interference of emotional stimuli or reducing the duration of time in which individuals attend to emotion stimuli (Brown et al., 2013; Teper & Inzlicht, 2013). This same process may well be occurring in children, but may not yet be well-developed enough to influence the evolutionarily hard-wired prominence of pain in capturing attention (Crombez et al., 2005). Finally, task performance on the 1-back significantly predicted distraction effectiveness for moderately painful heat stimuli. The effect size of the relation between task performance and pain unpleasantness change was larger than those found in adult studies (Buhle & Wager, 2010). Additional research should explore distraction task performance as a moderator of the effectiveness of distraction.

This study examined pain intensity and pain unpleasantness ratings separately to capture valuable information about sensory and affective processing of pain in children. Results are consistent with previous research that has found higher pain intensity ratings compared to pain unpleasantness ratings (Lu et al., 2007). However, it is worth noting that in the present study, and the study by Lu and colleagues, pain unpleasantness was rated following pain intensity for all trials. So, it is possible that the relatively lower unpleasantness ratings are an artefact of the order in which the ratings were obtained.

In addition to those noted above, there are several limitations to the current study. For ethical reasons (to avoid undue coercion), the consent and experimental scripts used during the experiment emphasized the participant's ability to discontinue at any time, in addition to stating that all heat levels delivered would be safe and would not cause actual damage to skin. It is likely that adding a high level of perceived control reduced the overall noxious and threatening aspects of pain, thereby reducing the potential influence of variables that influence pain affect, such as mindfulness. Additionally, although behavioural data suggest that the control task required directed attention, the possibility that participants were responding randomly cannot be ruled out. Future research should consider either using formal eye gaze data or measurement of keystrokes to evaluate for random responding or use a control task that requires discrimination of stimuli that could not be executed randomly.

This study offers several notable clinical implications to the literature. This study suggests that distraction tasks used in clinical settings for medical procedures (e.g., immunizations) will likely be more effective in reducing the general noxious aspects of pain for children if they are determined to have higher cognitive load, and may be especially beneficial for medical procedures that are associated with moderate levels of pain. By selecting the most effective distractors, it is anticipated that distraction will produce greater effects for children undergoing painful medical procedures. Improved pain care in clinical settings is further important because it is known to correlate with a wealth of critical factors including increased healthcare use in adulthood, better parent and family coping over time, and lower provider and healthcare costs (Cohen et al., 1997; Kazak et al., 1995; Luthy et al., 2009; Pate et al., 1996).

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CONFLICT OF INTEREST

The authors have no conflicts of interest to disclose.

AUTHOR CONTRIBUTIONS

The project was conceptualized by Dr. Gaultney as part of the requirements for her doctoral dissertation. Dr. Dahlquist served as Dr. Gaultney's committee chair and supervised all phases of the study including study design, data analysis and interpretation, and manuscript preparation. Dr. Quiton also actively served on Dr. Gaultney's dissertation committee, provided methodological support and participated in manuscript preparation.

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ENDNOTE

[1] We were unable to calculate the minimal detectable effect sizes for serial mediation, as we are not aware of software that is capable of this analysis with three mediators.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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